

Human-Actionable Checklist

Pre-Deployment Checklist

Before deploying a multiagent system in production, verify the following. Figure 1 provides a visual overview of the deployment phases and their associated verification checkpoints.

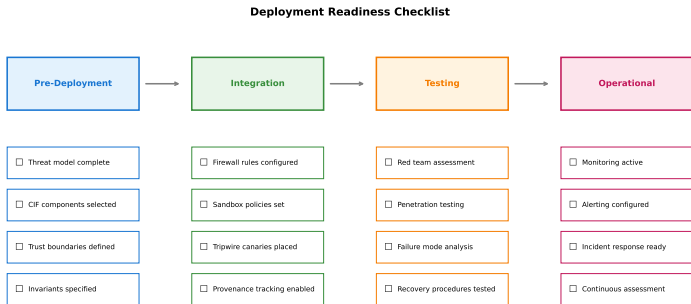


Figure 1: Deployment Readiness Checklist. The cognitive security deployment lifecycle consists of four phases: Pre-Deployment (threat model completion, CIF component selection, trust boundary definition), Integration (firewall configuration, sandbox policies, tripwire placement), Testing (red team assessment, penetration testing, failure mode analysis),

Operational Checklist (Daily/Weekly)

Daily Monitoring

- ☐ **Review drift alerts:** Check for unusual belief changes
- ☐ **Verify tripwire integrity:** Confirm canary beliefs unchanged
- ☐ **Check trust metrics:** Monitor for unexpected trust score changes
- ☐ **Review failed consensus:** Investigate any Byzantine fault indications

Weekly Review

- ☐ **Analyze attack patterns:** Review blocked injection attempts
- ☐ **Audit delegation chains:** Check for unusual delegation patterns
- ☐ **Verify invariant compliance:** Confirm no invariant violations
- ☐ **Update threat intel:** Incorporate new attack techniques into defenses

Incident Response Checklist

When a cognitive attack is suspected:

Immediate Actions (First 15 Minutes)

- ☐ **Preserve evidence:** Capture current cognitive state before any changes
- ☐ **Assess scope:** Identify which agents and beliefs may be affected
- ☐ **Contain spread:** Isolate affected agents from propagating beliefs
- ☐ **Notify stakeholders:** Alert security team and relevant operators

Investigation (First Hour)

- ☐ **Trace provenance:** Follow belief origins to identify injection point
- ☐ **Identify attack vector:** Determine how adversarial content entered
- ☐ **Assess impact:** Evaluate what decisions were influenced
- ☐ **Check for persistence:** Verify attack doesn't survive agent

Configuration Quick Reference

Trust Calculus Parameters

Parameter	Recommended Value	When to Adjust
Base weight ()	0.3	Increase for stable architecture
Reputation weight ()	0.4	Decrease for new deployments
Context weight ()	0.3	Increase for specialized tasks
Decay factor ()	0.9	Decrease for security-critical

Firewall Thresholds

Threshold	Recommended Value	Risk Trade-off
Accept threshold	0.3	Lower = more strict, more false positives
Reject threshold	0.7	Higher = more permissive, more risk
Quarantine range	0.3-0.7	Narrower = faster decisions, less nuance